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Module 12: Installation, Storage, and Compute with Windows Server

1. What two options are provided in the Type of Installation window during Windows Server 2016 installation?

1. Upgrade

- Keeps existing files, settings, and applications.
- Used when upgrading from an older Windows Server version.

2. Custom (Install Windows Only)

- Performs a fresh installation.
- Deletes old settings and installs a new copy of Windows Server.

2. How to Configure Server Step by Step?

1. Install Windows Server.
2. Set Administrator password.
3. Configure server name.
4. Assign a static IP address.
5. Configure DNS settings.
6. Join the server to a domain (if required).
7. Install required server roles and features.
8. Configure storage and network settings.
9. Enable security updates and firewall.
10. Verify server functionality.

3. What are the Pre-Installation Tasks?

Pre-installation tasks are activities performed before installing Windows Server.

1. Check hardware requirements.
2. Verify software compatibility.
3. Back up important data.
4. Plan disk partitions.
5. Prepare installation media (DVD/USB).
6. Configure BIOS/UEFI settings.
7. Ensure network connectivity if needed.

4. What are the Post-Installation Tasks?

Post-installation tasks are performed after installing Windows Server.

1. Change administrator password.
2. Configure IP address and DNS.
3. Rename the server.
4. Install updates and patches.
5. Install required roles and features.
6. Configure firewall and security settings.
7. Create users and groups.
8. Verify system performance.

5. What is the Standard Upgrade Path for Windows Server?

The standard upgrade path is upgrading from an older version to the next supported version.

Example:

- Windows Server 2008 R2 → Windows Server 2012 R2
- Windows Server 2012 R2 → Windows Server 2016
- Windows Server 2016 → Windows Server 2019
- Windows Server 2019 → Windows Server 2022

Direct upgrades across many versions are generally not supported.

6. What is the Physical Structure of Active Directory (AD)?

The physical structure of AD defines how network resources are physically connected.

It consists of:

1. **Domain Controllers (DCs)** – Servers storing the AD database.
2. **Sites** – Represent physical locations connected through networks.
3. **Subnets** – Network segments associated with sites.

7. What are the Logical Components of Active Directory?

The logical structure organizes resources logically rather than physically.

Components are:

1. **Domain**
2. **Organizational Unit (OU)**
3. **Tree**
4. **Forest**

5. **Objects** (Users, Computers, Printers, Groups)

8. What is the Full Form of LDAP?

LDAP = Lightweight Directory Access Protocol

It is a protocol used to access and manage directory services such as Active Directory.

9. What is the Location of the AD Database?

The Active Directory database is stored in the file:

NTDS.dit

Default location:

C:\Windows\NTDS\NTDS.dit

10. What is Child DC?

Child DC (Child Domain Controller) is a Domain Controller that belongs to a child domain within a parent domain structure.

Example:

```
company.com
|
└─ sales.company.com
```

Here, **sales.company.com** is the child domain, and its Domain Controller is called a Child DC.

11. Explain the Term Forest in AD

A **Forest** is the highest-level structure in Active Directory.

It is a collection of one or more domain trees that share a common schema, configuration, and global catalog.

Example

```
Forest
|
├─ company.com
|
└─ branch.com
```

Both domains belong to the same forest and can share resources securely.

Key Point

Forest = Collection of multiple domain trees sharing a common Active Directory structure.

12. What is Active Directory? Check all that apply.

- An open-source directory server
- A Windows-only implementation of a directory server
- Microsoft's implementation of a directory server

- An LDAP-compatible directory server
- ✓ Microsoft's implementation of a directory server
- ✓ A Windows-only implementation of a directory server
- ✓ An LDAP-compatible directory server
- ✗ An open-source directory server

13. When you create an Active Directory domain, what's the name of the default user account?

- ✓ Administrator

The Administrator account is created automatically and has full control over the domain.

14. AD Domain Provides Which Advantages? (Check all that apply)

- ✓ Centralized authentication
- ✓ More detailed logging
- ✓ Centralized management with GPOs
- ✗ Better performance

15. What are the Minimum Hardware Requirements for Installing Windows Server 2016?

- **Processor:** 1.4 GHz, 64-bit
- **RAM:** 512 MB (2 GB recommended for GUI)
- **Disk Space:** 32 GB
- **Network Adapter:** Gigabit Ethernet adapter
- **Display:** 1024 × 768 or higher

16. Explain the Different Editions of Windows Server 2016 and Their Features

1. Essentials Edition

- For small businesses.
- Supports up to 25 users and 50 devices.
- Simple management.

2. Standard Edition

- Suitable for physical or lightly virtualized environments.
- Supports up to 2 virtual machines.

3. Datacenter Edition

- For highly virtualized environments.
- Supports unlimited virtual machines.

- Includes advanced features such as Storage Spaces Direct and Shielded VMs.

17. Walk Through the Steps of Installing Windows Server 2016 Using GUI Mode

1. Boot from Windows Server 2016 DVD/USB.
2. Select language, time, and keyboard settings.
3. Click **Install Now**.
4. Enter product key (if required).
5. Select **Windows Server 2016 Standard/Datacenter (Desktop Experience)**.
6. Accept license terms.
7. Select **Custom Installation**.
8. Choose installation partition.
9. Click **Next**.
10. Wait for installation to complete.
11. Restart system.
12. Set Administrator password and log in.

18. Describe the Steps for Installing Windows Server 2016 in Server Core Mode

1. Boot from installation media.
2. Select language settings.
3. Click **Install Now**.
4. Select **Windows Server 2016 Standard/Datacenter** (without Desktop Experience).
5. Accept license agreement.
6. Choose Custom Installation.
7. Select partition and continue.
8. Complete installation.
9. Set Administrator password.
10. Configure server using command-line tools or PowerShell.

19. How Do You Configure Network Settings During Windows Server 2016 Installation?

1. Open **Network and Sharing Center**.
2. Click **Change Adapter Settings**.
3. Select network adapter.
4. Open **Internet Protocol Version 4 (TCP/IPv4)**.
5. Enter:

- IP Address
 - Subnet Mask
 - Default Gateway
 - DNS Server
6. Save settings and test connectivity.

20. Explain the Process of Promoting a Windows Server to a Domain Controller

1. Open **Server Manager**.
2. Click **Add Roles and Features**.
3. Install **Active Directory Domain Services (AD DS)**.
4. After installation, click **Promote this server to a domain controller**.
5. Select:
 - Add a new forest, or
 - Add a domain controller to an existing domain.
6. Enter domain details.
7. Set Directory Services Restore Mode (DSRM) password.
8. Complete wizard and restart the server.

21. Discuss the Steps Involved in Upgrading from a Previous Version of Windows Server to Windows Server 2016

1. Verify upgrade compatibility.
2. Back up important data.
3. Insert Windows Server 2016 installation media.
4. Run **Setup.exe**.
5. Select **Upgrade** option.
6. Choose edition to install.
7. Accept license terms.
8. Complete compatibility checks.
9. Start upgrade process.
10. Restart system and verify services.

22. What is Active Directory Domain Services (AD DS), and What are Its Key Components?

Definition

AD DS (Active Directory Domain Services) is a Microsoft service that stores and manages information about users, computers, groups, and network resources in a domain.

Key Components

1. Domain
2. Domain Controller
3. Organizational Unit (OU)
4. Tree
5. Forest
6. Global Catalog
7. Schema

23. How Do You Create a New Active Directory User Account in Windows Server?

1. Open **Active Directory Users and Computers**.
2. Select the desired Organizational Unit (OU).
3. Right-click and select **New → User**.
4. Enter:
 - First Name
 - Last Name
 - Username
5. Click **Next**.
6. Set password.
7. Select password options.
8. Click **Finish**.

The new user account is created successfully.

24. Explain the Process of Creating and Managing Group Policy Objects (GPOs) in Windows Server 2016/2019

1. Open **Group Policy Management Console (GPMC)**.
2. Right-click the domain or OU.
3. Select **Create a GPO in this domain and Link it here**.
4. Enter a GPO name.
5. Right-click the GPO and select **Edit**.
6. Configure required settings.
7. Save changes.
8. Apply policies using:

gpupdate /force

GPOs are used to centrally manage users and computers.

25. What are Organizational Units (OUs) in Active Directory, and How Do You Use Them?

Definition

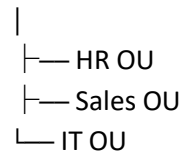
An **Organizational Unit (OU)** is a container in Active Directory used to organize users, groups, and computers.

Uses

- Organize resources logically.
- Apply Group Policies.
- Delegate administrative control.
- Simplify management.

Example

Company



26. Describe the Process of Delegating Administrative Privileges in Active Directory

1. Open **Active Directory Users and Computers**.
2. Right-click the required OU.
3. Select **Delegate Control**.
4. Click **Next**.
5. Add users or groups.
6. Select tasks to delegate.
7. Complete the wizard.

Advantages

- Reduces workload of domain administrators.
- Improves security.
- Allows department-level administration.
- Provides controlled access to resources.